

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (EE) Examination

Dec 2012

EE709 Distributed Generation

Date: 13.12.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Which are the various distributed generation technologies used in power system network? 07
Explain any three of them

Q - 2 (a) What are the reasons for adopting distributed generation in power system network? Discuss 07
any two of them in details

(b) Discuss the various drivers for the distributed generation growth in power system 07

OR

(b) Define Islanding? When it occurs? What are the various issues of islanding? 07

Q - 3 Attempt any two 14

(1) What are the various problems due to DG on grounding?

(2) Explain the photovoltaic effect and materials overview

(3) Discuss the challenges for the increased penetration of distributed generation

SECTION - II

Q - 4 Discuss the voltage level related connection cost for distributed generation system 07

Q - 5 (a) Discuss the PV system efficiency and basic design of PV system 07

(b) Explain the Deep Vs Shallow connection charges for distributed generation system by giving 07
an appropriate example

OR

(b) Explain following methods of islanding 07

(i) Remote detection technique

(ii) Local detection technique

Q - 6 Attempt any two 14

1. What is the impact on short circuit level due to incorporation of distributed generation

2. Discuss the factors that affects the cost of PV system

3. Discuss the procedure to allocate the losses in distributed power system
